

Replacing the body mass index with the sagittal abdominal diameter (abdominal height)

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Abstract

The body mass index (BMI; weight/height²), providing no information about the relative size of any adipose tissue depots, may accordingly misclassify degrees of cardiometabolic risk. However, in supine persons the abdominal height above the exam table (the sagittal abdominal diameter, SAD) is associated preferentially with the accumulation of visceral fat. Since visceral fat is a marker of insulin resistance, type 2 diabetes, coronary heart disease, and hypertension, SAD could contribute to the estimation of generalized cardiometabolic risk. The SAD has been measured inexpensively by a sliding-beam caliper in small studies and in the US National Health and Nutrition Examination Survey (NHANES). Cross-sectional models found that the SAD/height ratio (SADHtR) is more strongly associated than the waist circumference/height ratio or BMI with intermediary predictors of cardiometabolic disease. Prospective studies are needed, however, to demonstrate how well SAD or SADHtR might predict major disease outcomes or all-cause mortality.

In June 2023, the American Medical Association (AMA) adopted a report clarifying the role of the body mass index (BMI; weight in kilograms divided by height in meters squared) in clinical settings. Medical practice stands to benefit greatly from a practical adiposity indicator that could be collected inexpensively in most clinical settings. A simple estimator of generalized cardiometabolic risk, if validated through prospective studies, could enhance practice efficiency while reducing the overall costs of chronic-disease management. The AMA did not endorse any adiposity indicator as a replacement for BMI.

The AMA critique of BMI recognized 11 other indicators that can also describe excessive adiposity in adults. Nine of these indicators were substantially more complex and expensive than BMI and thus are unlikely to gain acceptance in typical clinical practice or epidemiological surveys. In this Perspective article I will point out the existence of a simple, but underappreciated, adiposity indicator that could serve where BMI has fallen short.

The AMA's report emphasized that the accumulation of visceral fat is a strong predictor of insulin resistance, type 2 diabetes, coronary heart disease, and hypertension. The standing waist circumference (WC) and waist-hip ratio (WHR) of circumferences were mentioned as

two relatively simple adiposity indicators that address body-fat distribution. Compared to multi-slice imaging by computed tomography (CT) or magnetic resonance imaging (MRI), WC is a quicker and much less expensive indicator of the abdominal adiposity burden. However, WC incorporates not only visceral fat but also subcutaneous abdominal fat, which tends to be neutral toward predicting adverse outcomes [1]. The WHR is a slightly more complex risk indicator than WC alone without consistent benefits for identifying cardiometabolic risk. It should be noted that there are several distinct protocols for accurately measuring the standing circumferences at either the waist or the hip. Any of these measurement protocols requires careful instruction and disciplined application. No protocol for measuring these two circumferences has yet been universally accepted.

Overlooked by the recent AMA report is another low-cost measurement that estimates visceral fat accumulation. Swedish researchers 35 years ago recommended measuring the supine sagittal abdominal diameter (SAD; also called the "abdominal height") [2]. Their early work measured SAD on abdominal CT images obtained from 43 adults. The SAD values extracted from the image at the L4–5 interspace level were superior to the WC and the WHR for predicting

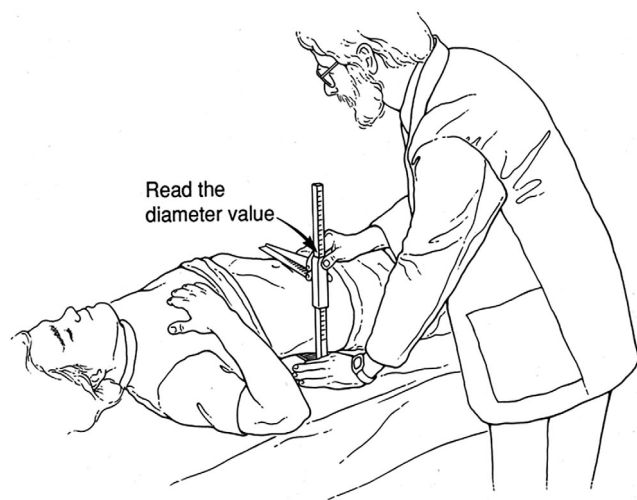


FIGURE 1 Measurement by sliding-beam caliper of the sagittal abdominal diameter on a supine participant. Drawing adapted with permission [10].

the visceral fat volume. Because abdominal subcutaneous fat falls away toward the sides of a supine belly, subcutaneous fat contributes relatively little to the variation in SAD. The visceral fat, however, is constrained by muscle and bone to remain near the midline. In larger, more recent studies employing MRI or CT images, SAD was confirmed to be correlated more strongly than WC with the amount of visceral fat [3, 4].

Measurement of the supine SAD is simpler and less expensive with the use of a portable, sliding-beam caliper (Figure 1). Following research reports from several countries, an abdominal caliper was adopted in 2011 by the US National Health and Nutrition Examination Survey (NHANES) for application in its nationally representative population samples. Backed up by 6 years of survey experience in the field, a clear protocol for SAD measurement by a sliding-beam caliper was published in the NHANES Anthropometry Procedures Manual [5]. Normative reference values have been published for both the SAD [6] and the SAD/height ratio (SADHtR) [7]. Cross-sectional, multiply-adjusted models have demonstrated that the SADHtR was more strongly associated than the WC/height ratio or BMI with fasting triglycerides, fasting insulin, and three proxy estimators of insulin resistance [8].

Triglycerides, insulin, and insulin resistance, however, are only intermediary outcomes. Urgently needed are well-designed projects to compare the abilities of simple abdominal measurements, including SAD, to predict adiposity-related major adverse events. One prospective cohort study reported in multiply-adjusted models that SAD predicted incident, ischemic cardiovascular events more strongly than BMI or WC; the SADHtR was a slightly stronger predictor than the uncorrected SAD [9]. This sole published report, however, was restricted to adults who were aged 60 years old at baseline, and the supine SAD was measured using a bedside ruler and a water level. Future longitudinal studies must include a wider range of ages and

SAD values measured by a standardized caliper instead of adapted carpentry tools.

The NHANES data sets, freely accessible online, could support the need for suitable prospective cohorts. NHANES provides not only high-quality anthropometric measurements but also an extraordinary collection of clinical, laboratory, dietary, and other variables collected simultaneously from survey participants. Well-prepared analysts can link these data to other US files, including the National Death Index. Thus, since SAD was obtained by NHANES in 2011–2016, it may soon be possible to compare SAD with other adiposity indicators for predicting all-cause mortality during at least 5 years of follow-up experience.

Despite decades of recommendation, it is still rare to find WC implemented in medical practice. Conscientious health professionals may be unenthusiastic about adopting a measurement taken at anatomic landmarks that are not universally accepted. Or they may regret the imprecision of standing WC values that may have arisen from a conflict between minimizing tension on the measuring tape and the tape's tendency to droop below the horizontal plane.

The recognition of BMI's inadequacies by the AMA and others challenges us to advocate for a reproducible, effective, but simple, anthropometric replacement. A demonstration of epidemiologic efficacy may still be an insufficient rationale for practitioners to change how they assess generalized, adiposity-related cardiometabolic risk. Those of us associated with large health care systems might consider different research agendas. We could use economic analyses to study the potential cost savings attributable to patient care guided by improved anthropometry at the patients' intake and during their subsequent management. Such research might be the most influential approach to advocacy in our era of rising health care costs. Whether SAD, WC, or some other adiposity indicator turns out to provide the greatest cost-effectiveness, we will have advanced the patient-care goals of obesity science.○

DISCLOSURE

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